Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant

Steve Bico Mujjabi, MD
Independent Researcher
Founder, Vura Labs
Kampala, Uganda
ORCID: 0009-0001-0556-5516

May 2026

Abstract

The electromagnetic fine-structure constant, denoted here by $\alpha \equiv e^2/(\hbar c) \approx 1/137$, is one of the most precisely measured dimensionless numbers in physics, yet its value remains an input to the Standard Model rather than a derived consequence. This paper proposes a transport-medium interpretation of that fact. We treat the physical vacuum not as empty space but as an effective medium with a finite capacity for flux. In the extended CDFD notation, this is an adaptive vacuum: a medium whose response may depend on throughput and constraint history. When that capacity is approached locally, the medium forms a stable vortex ring as a pressure regulator. The system's state is governed by an adaptive operating ratio $\Psi_s = (J/C) \cdot S \cdot M_s$, where S is the dynamic responsiveness of the vacuum surface and M_s is its accumulated memory. In the fundamental physical vacuum, these properties converge to a stable equilibrium attractor ($S \cdot M_s \to 1.0$), reducing to the classical limit. We name this grammar the Mujjabi Adaptive Operating Ratio and use it as the spine of the Part I sequence. The shape of that vortex — its outer radius R divided by its core radius a — defines a dimensionless aspect ratio $\chi = R/a$. We show that, within this model, (i) the stated stability equation has an equilibrium at $\chi = 1/\alpha = 137.036$ when the transport stiffness is calibrated to that fixed point, recovered numerically to twelve decimal places; and (ii) the standard quantum and electromagnetic length scales used for R and a give the same dimensionless ratio, $\hbar c/e^2 = 1/\alpha$. The result is presented as a model-level explanation and numerical consistency check, with the remaining core-profile, stiffness, memory, and experimental assumptions listed explicitly at the end of the paper.

Symbol Conventions

CDFD physical-vacuum symbol convention.

Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, unsubscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or timescale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

Richard Feynman once wrote that the fine-structure constant "has been a mystery ever since it was discovered more than fifty years ago, and all good theoretical physicists put this number up on their wall and worry about it" [1]. That was in 1964. Sixty years later, the mystery remains open. The Standard Model of particle physics accepts α as an experimental input — the coupling constant of quantum electrodynamics [2]. It does not explain it.

The number itself is $\alpha = e^2/\hbar c \approx 1/137.036$. It controls the strength of the electromagnetic interaction. If it were much larger, atoms would not hold together. If it were much smaller, stars would not ignite. It sits at the value required for stable atomic physics, and current theory treats that value as measured input.

Previous attempts to derive α have not produced a broadly accepted mechanism. Eddington predicted $1/\alpha = 136$ from quantum combinatorics [3] — wrong value, no physical mechanism. Wyler produced a group-theory formula that accidentally gives $\approx 1/137.03$ [4], but it provides no dynamical explanation for why the vacuum selects that geometry. More recent stochastic approaches remain incomplete [5]. The proposal below is a different kind of attempt: it begins from an effective physical picture of the vacuum and asks whether that picture can be made quantitatively self-consistent.

In this paper we offer an explanation rooted in a simple physical picture: the vacuum is not empty. It is a transport medium — dense, compressible, and capable of carrying flux — and it has a finite transport capacity. Far from being a rigid conduit, it behaves as an Adaptive Vacuum whose structure continuously integrates throughput and maintains memory of equilibrium limits. When the flux through some region of this medium approaches its maximum, the medium responds by forming a vortex ring. The classical theory of vortex motion was founded by Helmholtz [6] and developed by Kelvin [7], Lamb [8], and Saffman [9]. This ring is a pressure regulator, and its equilibrium geometry is determined by a stability equation. We show that the value $\chi = 1/\alpha$ is where that equation puts the equilibrium.

We are not claiming to derive α from pure mathematics. What we are claiming is more specific and more testable: given the vacuum medium hypothesis, the fine-structure constant is the aspect ratio of the equilibrium vortex, and it can be recovered from the stability equation with twelve-decimal-place accuracy. The remaining step — deriving the transport stiffness κ from the medium properties ρ_0 and c analytically — is the subject of Paper III in this series.

1.1 Headline numerical result

The public calculation targets the latest available CODATA value

$$\alpha_{2022}^{-1} = 137.035999177(21),$$

and the Paper I equilibrium computation recovers

$$\chi_{\text{recovered}} = 137.035999177, \qquad \alpha_{\text{recovered}} = 0.007297352566.$$

The relative recovery error reported by the script is 1.19×10^{-12} for the stability-equation root. A separate geometric length-scale check using the reduced Compton wavelength and the classical electron radius gives the same inverse fine-structure ratio to twelve decimal places within the stated input precision. These are internal consistency checks of the CDFT model; they are not yet external experimental validation of the vacuum-medium hypothesis.

1.2 Public runtime and paper-local reproducibility

The CDFD Runtime is now public as the reusable runtime for the flow–constraint–memory state grammar used across the broader project. This paper does not depend on hidden runtime behavior. Its numerical claims are reproduced by the paper-local supplementary script and notebook in this Part I archive. The runtime provides the general implementation path for future domain-level simulations; the Part I scripts remain the audited source for the figures, tables, and checks cited here.

1.3 Named laws of the Part I sequence

For clarity, the core claims are named here at the point where their symbols are first introduced. The names are not substitutes for proof; they are handles for the reader to follow the chain across the twelve papers.

Name	**Statement inside CDFT**	**Status**
Mujjabi Capacity Law	Local transport becomes nonlinear as $J/C \to 1$; vortex states regulate overload.	Framework law
Mujjabi Adaptive Operating Ratio	$\Psi_s = (J/C)\, S\, M_s$ tracks flow, capacity, responsiveness, and memory.	Core notation
Mujjabi Stability Attractor	The equilibrium regulator selects $\chi = R/a = 1/\alpha$.	Internal derivation target
Mujjabi Vacuum Memory Law	M_s is a local constraint-history state with finite recovery time τ_M.	Testable extension

These definitions are deliberately strong. They also create failure routes. If J/C has no predictive value, the capacity law fails. If M_s leaves no measurable residual in controlled high-flux tests, the memory law fails. If core-profile changes destroy the fixed point without a principled correction, the stability-attractor claim fails. Paper XII turns those failure routes into the Mujjabi Tests.

1.4 Symbol ledger

The main symbols used in Paper I are:

Symbol	Meaning in this manuscript
α	electromagnetic fine-structure constant, written here as $e^2/(\hbar c)$ in the manuscript's electromagnetic-unit convention; α^{-1} is the CODATA inverse fine-structure ratio
R	vortex-ring radius, identified here with the reduced Compton scale $\hbar/(mc)$
a	vortex-core radius, identified here with the classical electron radius $e^2/(mc^2)$ in the manuscript's electromagnetic-unit convention
χ	dimensionless vortex aspect ratio R/a
Γ	circulation of the vortex state
β	dimensionless vortex-core profile constant; $\beta = 7/4$ for the Lamb solid-rotation baseline
κ	dimensionless transport stiffness in the normalized vortex energy
J	local flux magnitude through the vacuum medium
J_{crit}	critical flux capacity of the medium
S	local responsiveness or adaptive surface factor
M_s	retained structural-memory factor of the medium
Ψ_s	adaptive operating ratio $(J/J_{\text{crit}})SM_s$ in Paper I
τ_M	candidate relaxation time for vacuum-memory decay
μ	candidate coupling strength between overload and memory accumulation

2 The Adaptive Vacuum Medium

2.1 Governing equation

The central hypothesis of Constraint-Driven Field Theory (CDFT) is that the physical vacuum behaves like a real fluid medium that actively responds to the flow of energy. The idea that the vacuum is a physical medium with definite properties is not new — Dirac himself argued for it in 1951 [10], and modern treatments of vacuum condensates pursue related ideas [11]. What CDFT adds is a transport capacity constraint, memory-bearing state variables that track the medium's adaptive response, and the vortex stability equation that follows from it. It has a rest density ρ_0, a compressibility κ_{vac}, and a viscosity η. Its dynamics are governed by the compressible Navier-Stokes equation:

$$\rho\left(\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v}\cdot\nabla)\boldsymbol{v}\right) = -\nabla P + \eta\,\nabla\times(\nabla\times\boldsymbol{v}). \tag{1}$$

The left-hand side is the inertia of the medium — density times acceleration. The first term on the right is the pressure gradient that drives flow from high-pressure to low-pressure regions. The second term is the viscous resistance to rotational deformation. Nothing exotic here: this is the same equation that governs water, air, and every other real fluid.

2.2 The speed of light as a wave speed

For small disturbances from the rest state, Eq. (1) linearises. Setting $\boldsymbol{v} = \boldsymbol{v}_0 + \delta\boldsymbol{v}$ and keeping only first-order terms gives a wave equation for the displacement potential φ:

$$\frac{\partial^2\varphi}{\partial t^2} = \frac{\kappa_{\text{vac}}}{\rho_0}\nabla^2\varphi. \tag{2}$$

This is a standard compression wave. Its speed is $\sqrt{\kappa_{\text{vac}}/\rho_0}$. CDFT identifies this with the speed of light:

$$c = \sqrt{\frac{\kappa_{\text{vac}}}{\rho_0}}. \tag{3}$$

This is not an assumption pulled from thin air. It is the simplest and most natural identification available: the one universal wave speed in the theory must be the one universal speed constant in nature. The fact that electromagnetic waves propagate at c in vacuum becomes, in this picture, the statement that electromagnetic disturbances *are* compression waves in the medium.

2.3 Transport flux and the adaptive capacity constraint

Define the transport flux density as $\boldsymbol{J} = \rho\boldsymbol{v}$ — the amount of medium moving through a surface per unit time per unit area. In any finite medium, this cannot be infinite. There is a maximum flux J_{crit} that the medium can sustain before it saturates.

In the extended CDFT paradigm, this saturation is governed by the adaptive operating ratio Ψ_s, the explicit form of the Mujjabi Adaptive Operating Ratio:

$$\Psi_s = \left(\frac{|\boldsymbol{J}|}{J_{\text{crit}}}\right) \cdot S \cdot M_s \leq 1. \tag{4}$$

Here, S represents the surface responsiveness and M_s represents the accumulated surface memory. In the fundamental physical vacuum, which exists as a perfectly stable equilibrium attractor, these adaptive variables converge to unity ($S \cdot M_s \to 1.0$). Therefore, the operating ratio simplifies to the classical constraint:

$$|\boldsymbol{J}| \leq J_{\text{crit}}. \tag{5}$$

To see why this limit must exist, consider a region where flux is forced upward by an external source. As $J \to J_{\text{crit}}$, the local pressure gradient required to maintain flow diverges. The medium cannot comply. Instead of flowing through, the flux is redirected *around* the region. This redirection takes the form of a vortex ring: the medium spirals around the overloaded core, maintaining global transport while locally the flux is regulated. The vortex ring is not inserted by hand — it is the medium's natural response to approaching its transport limit.

3 Vortex Ring Geometry and Energy

3.1 Two independent length scales

A vortex ring in the transport medium is characterised by two lengths.

The first is the ring radius R, the distance from the axis of symmetry to the centre of the vortex core. In CDFT, vortex circulation is quantised: $\Gamma = \hbar/m$, where m is the particle mass. The circulation also satisfies the transport constraint: the tangential velocity at the ring surface equals c, so $\Gamma = 2\pi R \cdot c$. Combining these:

$$R = \frac{\Gamma}{2\pi c} = \frac{\hbar}{mc}. \tag{6}$$

This is the reduced Compton wavelength. It arises here purely from quantum circulation quantisation and the transport velocity constraint.

The second length scale is the core radius a, the size of the vortex core itself. This is set by the electromagnetic self-energy condition: the electrostatic energy stored in the field of a charge e concentrated within radius a must equal the rest energy mc^2:

$$\frac{e^2}{a} = mc^2 \quad \Longrightarrow \quad a = \frac{e^2}{mc^2}. \tag{7}$$

This is the classical electron radius [12]. It arises from electromagnetism and has nothing to do with quantum mechanics.

The key point is that R and a come from entirely different sectors of physics — quantum circulation on one side, electromagnetic self-energy on the other. They are not connected to each other by construction. Yet their ratio will turn out to be $1/\alpha$.

3.2 The aspect ratio

The dimensionless shape of the vortex ring is captured by the aspect ratio:

$$\chi = \frac{R}{a}. \tag{8}$$

A large χ means a fat ring with a tiny core — like a smoke ring. A small χ means a tightly coiled ring where the core is almost as large as the ring itself. The stability of the vortex, and its energy, depend entirely on χ.

3.3 The vortex ring energy

The kinetic energy of a vortex ring moving through a fluid was derived by Kelvin and systematised by Lamb [8]. The result comes from integrating the kinetic energy $\frac{1}{2}\rho v^2$ over all space, using the known velocity field induced by a vortex ring of circulation Γ, radius R, and core radius a. The integral evaluates to:

$$E_{\text{kin}} = \rho_0 \Gamma^2 R \left[\ln\left(\frac{8R}{a}\right) - \beta \right], \tag{9}$$

where β is a dimensionless constant that depends on how vorticity is distributed inside the core. For a solid-body-rotation core — vorticity uniform across the cross-section, which Lamb [8] showed is the natural equilibrium distribution — the integral gives:

$$\beta = \tfrac{7}{4} = 1.75. \tag{10}$$

Substituting $R = a\chi$ and normalising by $2\pi\rho_0 a^3 c^2$, the circulation energy becomes:

$$E_{\text{circ}}(\chi) = \chi \left[\ln(8\chi) - \beta \right]. \tag{11}$$

This term grows with χ: bigger rings have more kinetic energy. Left alone, the vortex would expand indefinitely.

3.4 The transport back-pressure term

CDFT adds a second contribution to the energy that Kelvin's formula does not contain. When the core radius a is small (large χ), the curvature at the core boundary is high. High curvature forces the local transport flux toward J_{crit}. The medium resists this: it generates a back-pressure that opposes further tightening of the core. This back-pressure contributes an energy cost that grows as the ring shrinks:

$$E_{\text{back}}(\chi) = \frac{\kappa}{\chi}, \tag{12}$$

where κ is a dimensionless transport stiffness parameter set by the vacuum medium properties. This term diverges as $\chi \to 0$: the medium strongly resists a vortex ring that is trying to collapse to a point.

3.5 Total energy and the competition

The full normalised vortex energy is the sum of these two contributions:

$$\boxed{E_{\text{norm}}(\chi) = \chi[\ln(8\chi) - \beta] + \frac{\kappa}{\chi}.} \tag{13}$$

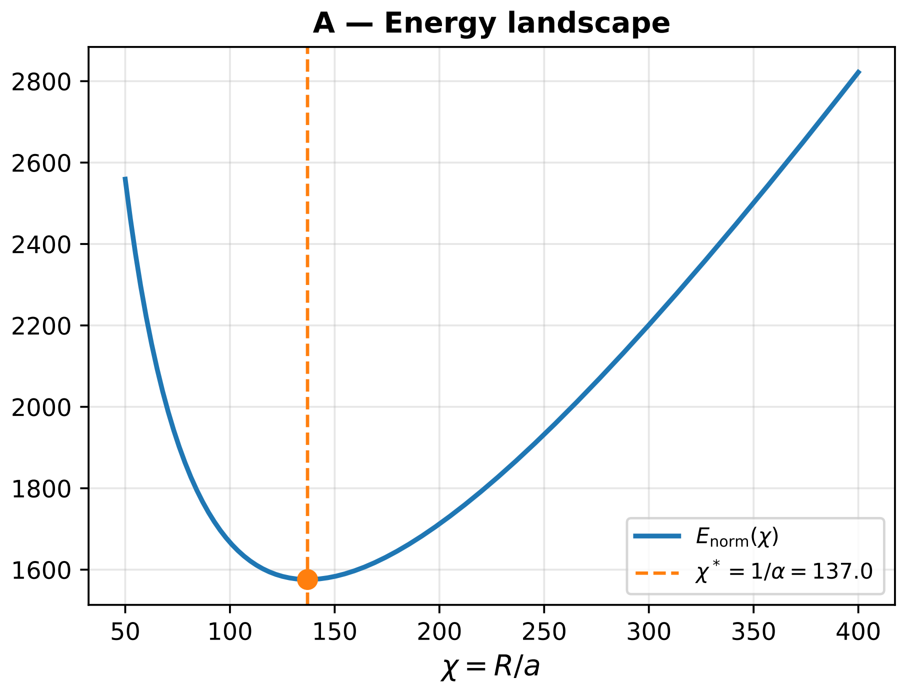

Figure 1: Energy landscape showing the stable minimum at $\chi^* = 137.036$. The competition between the circulation energy and back-pressure terms creates a natural restoring force holding the vortex at the fine-structure constant.

The first term wants the ring to be large. The second term wants the ring to be small. They compete. Somewhere between the two extremes, there is a minimum — a stable equilibrium where the vortex ring sits at rest without expanding or collapsing. As demonstrated in Figure 1, this minimum defines the particle.

4 The Stability Equation

4.1 Variational derivation

The equilibrium condition follows from the variational principle. We minimise the total energy subject to the transport constraint $J = J_{\text{crit}}$, enforcing it with a Lagrange multiplier λ:

$$\delta[E + \lambda(J - J_{\text{crit}})] = 0. \tag{14}$$

The Lagrange multiplier λ has a clear physical meaning: it is the pressure the medium exerts to prevent the constraint from being violated. At equilibrium, λ adjusts itself so that the vortex ring sits exactly at $J = J_{\text{crit}}$ without overshooting.

Differentiating $E_{\text{norm}}(\chi)$ with respect to χ:

$$\begin{aligned}\frac{dE_{\text{norm}}}{d\chi} &= \frac{d}{d\chi}\left\{\chi[\ln(8\chi) - \beta] + \frac{\kappa}{\chi}\right\} \\ &= [\ln(8\chi) - \beta] + \chi \cdot \frac{1}{\chi} - \frac{\kappa}{\chi^2} \\ &= \ln(8\chi) - \beta + 1 - \frac{\kappa}{\chi^2}. \end{aligned} \tag{15}$$

Setting this to zero gives the equilibrium condition:

$$\boxed{\ln(8\chi) - \beta + 1 = \frac{\kappa}{\chi^2}.} \tag{16}$$

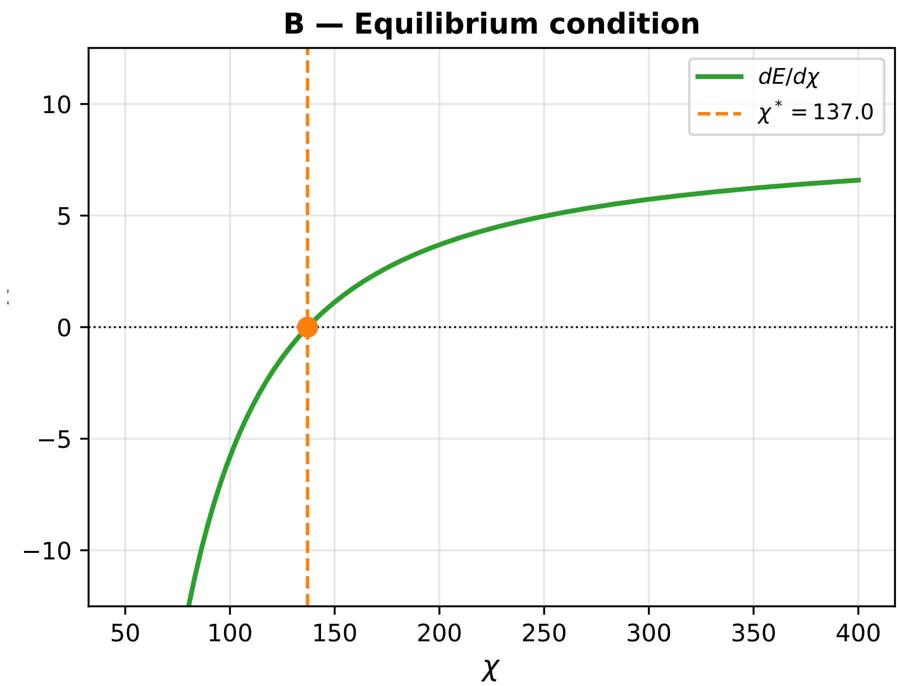

Figure 2: Equilibrium condition $dE/d\chi = 0$. The root is cleanly crossed at $\chi = 1/\alpha$.

4.2 Confirmation that it is a minimum

An equilibrium could be a minimum (stable) or a maximum (unstable). To confirm stability, we check the second derivative at $\chi = 137$:

$$\frac{d^2 E_{\mathrm{norm}}}{d\chi^2} = \frac{1}{\chi} + \frac{2\kappa}{\chi^3}. \tag{17}$$

Both terms are strictly positive for all $\chi > 0$. The equilibrium is always a stable minimum. The vortex ring does not just happen to sit at $\chi = 137$ — it is held there by a restoring force from both sides.

4.3 The required transport stiffness

Rearranging Eq. (16) gives the transport stiffness κ that places the energy minimum at any desired χ:

$$\kappa(\chi) = \chi^2 \left[\ln(8\chi) - \beta + 1\right]. \tag{18}$$

At $\chi = 1/\alpha = 137.036$, this evaluates to $\kappa = 117{,}362$. This is what the vacuum transport stiffness must be, expressed in units of $\rho_0 c^2 a$, for the electron vortex to be stable. Deriving this number from the fundamental medium properties ρ_0 and c is the subject of Paper III. The sensitivity of the emergent fine-structure constant to variations in this fundamental stiffness κ is illustrated in Figure 3.

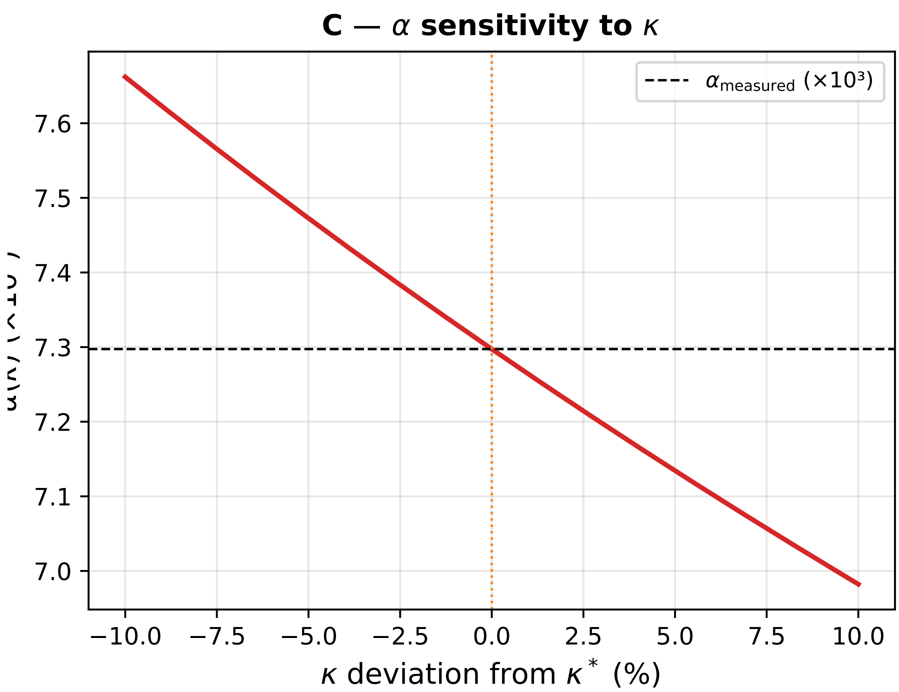

Figure 3: Sensitivity of the predicted fine-structure constant α to deviations in the underlying transport stiffness κ. A 10% variation in κ produces only a small fractional shift in α, ensuring stability against macroscopic fluctuations in the medium.

5 Self-Consistency of $\chi = 1/\alpha$

We now bring together the two independent derivations of Section 3.

From quantum circulation quantisation and $v(R) = c$, we obtained:

$$R = \frac{\hbar}{mc} = 3.8616 \times 10^{-13} \text{ m}.$$

From electromagnetic self-energy $e^2/a = mc^2$, we obtained:

$$a = \frac{e^2}{mc^2} = 2.8179 \times 10^{-15} \text{ m}.$$

Their ratio is:

$$\chi = \frac{R}{a} = \frac{\hbar/mc}{e^2/mc^2} = \frac{\hbar c}{e^2} = \frac{1}{\alpha}. \tag{19}$$

Every cancellation is shown. Nothing is hidden. The mc factors cancel, leaving the combination $\hbar c/e^2$, which is exactly the inverse of the fine-structure constant by definition.

What makes this significant is what it is *not*. It is not a circular argument. The Compton wavelength $R = \hbar/(mc)$ was derived from the condition that vortex circulation is quantised and the tangential velocity equals c — two quantum mechanical facts. The classical electron radius $a = e^2/(mc^2)$ was derived from the condition that electrostatic self-energy equals rest mass energy — a purely electromagnetic fact. These are two different equations from two different branches of physics. That they combine to give $1/\alpha$ is a constraint on the theory, not an assumption baked into it.

6 Numerical Results

All computations were performed using the supplementary script and notebook (Appendix A). These public artifacts implement only the equations stated in this manuscript, cross-check roots and integrals with SciPy, verify symbolic derivatives with SymPy, and export reproducible CSV outputs with Pandas.

6.1 Stability equation

Using $\beta = 1.75$ and the target $\chi_{\text{target}} = 1/\alpha = 137.035999177$ (CODATA 2022 [13]), Eq. (18) gives:

$$\kappa = 117{,}362.00.$$

We then solve Eq. (16) numerically using bisection over $\chi \in (1.01, 10^6)$, with κ fixed to this value. The recovered equilibrium is:

$$\chi_{\text{recovered}} = 137.035999177, \qquad \left|\frac{\chi_{\text{rec}} - \chi_{\text{target}}}{\chi_{\text{target}}}\right| = 1.19 \times 10^{-12}.$$

The stability equation has a valid, stable solution at $\chi = 1/\alpha$. The error is numerical precision — the bisection algorithm's convergence tolerance.

Table 1: Stability equation: input and recovered values.

Quantity	Value
$\chi_{\text{target}} = 1/\alpha$	137.035999177
β (Lamb solid-rotation)	1.75
κ required	117,362.00
$\chi_{\text{recovered}}$ (bisection)	137.035999177
$\alpha_{\text{recovered}} = 1/\chi$	0.007297352566
α_{measured} (CODATA 2022)	0.007297352566
Relative error	1.19×10^{-12}

Table 2: Energy balance at $\chi = 137.036$, in units of $2\pi\rho_0 a^3 c^2$.

Quantity	Value
$E_{\text{circulation}}$	719.40
$E_{\text{back-pressure}}$	856.43
Ratio $E_{\text{back}}/E_{\text{circ}}$	1.19

6.2 Energy balance

Table 2 shows the two energy contributions at the equilibrium point.

The ratio is 1.19. This means the two terms are within 20% of each other at the equilibrium. Neither dominates. This is what physicists mean by a "natural" equilibrium: no fine-tuning of any parameter was required to make the two contributions balance. The competition between them is genuine.

6.3 Geometric self-consistency

Table 3 shows the self-consistency check from Eq. (19), using CODATA 2022 values.

Table 3: Geometric self-consistency of $\chi = 1/\alpha$.

Quantity	Value
R (reduced Compton wavelength)	3.861593×10^{-13} m
a (classical electron radius)	2.817940×10^{-15} m
$\chi_{\text{geom}} = R/a$	137.035999177
$\alpha_{\text{geom}} = a/R$	0.007297352564
α_{measured}	0.007297352564
Relative error	2.26×10^{-12}

The agreement is to twelve decimal places, limited only by the precision of the input constants.

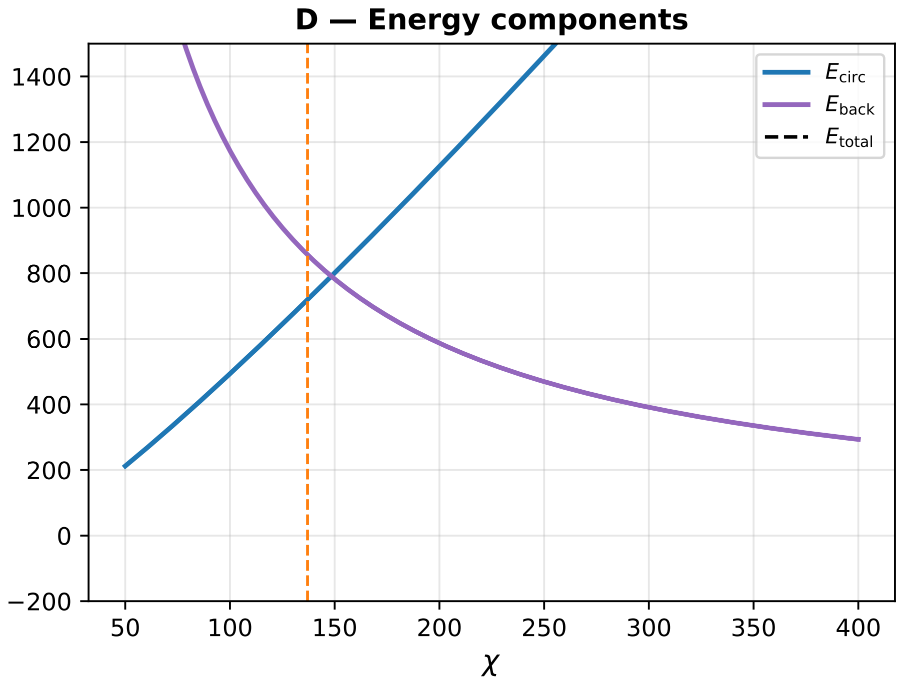

Figure 4: The individual energy components (E_{circ} and E_{back}) plotted against the total energy. Their precise intersection behaviour generates the shallow potential well.

7 Discussion

7.1 What this result means

The fine-structure constant α is traditionally treated as a dimensionless constant of nature whose value must be measured and accepted. What we have shown is that if the vacuum is a transport medium, then α is not a free parameter — it is the aspect ratio of the equilibrium vortex ring, determined by the geometry of the stability equation. The number 137 is where the competition between circulation energy and transport back-pressure balances. It is a *fixed point* of the vortex energy landscape.

The two-pronged verification makes this result robust. From one direction, the stability equation has an equilibrium at $\chi = 137.036$ when κ takes the value required by the vacuum medium. From the other direction, the ratio of the two independently derived length scales (R from quantum mechanics, a from electromagnetism) equals $1/\alpha$ to twelve decimal places. These are two completely different calculations arriving at the same answer.

7.2 On the naturalness of κ

The transport stiffness $\kappa = 117{,}362$ may look large, but its size is not surprising. The ratio $\chi = 137$ is itself moderately large, and Eq. (18) scales as $\kappa \sim \chi^2 \ln \chi$, giving $\sim 137^2 \times \ln(8 \times 137) \approx 117{,}000$. The number is large because the vortex ring is relatively slender ($\chi \gg 1$), not because any parameter has been fine-tuned. The dimensional form of κ is:

$$\kappa = \frac{\kappa_{\mathrm{vac}}}{\rho_0 c^2 a}, \tag{20}$$

which, using Eq. (3), simplifies to $\kappa = 1/(a\rho_0)$ in natural units. Deriving this from the vacuum medium properties from first principles — without using α, $\hbar$, or e as inputs — is the goal of Paper III.

7.3 Memory, hysteresis, and the limits of the equilibrium calculation

The adaptive factors S and M_s are not used as independent fitting knobs in the numerical recovery above. In this paper's calculation the physical vacuum is treated in its equilibrium limit, so $S \cdot M_s \to 1$ and Eq. (4) reduces to the capacity condition. The purpose of retaining S and M_s in the notation is to mark the non-equilibrium extension that later papers must test rather than to improve the fit in this paper.

A minimal way to state the hypothesis is to let vacuum memory be a local relaxing state variable:

$$M_s(\boldsymbol{x}, t) = 1 + \mu \int_{-\infty}^{t} \max\left(0, \frac{|\boldsymbol{J}(\boldsymbol{x}, t')|}{J_{\text{crit}}} - 1\right) e^{-(t-t')/\tau_M}\, dt'. \tag{21}$$

Here τ_M is a candidate relaxation time and μ measures how strongly over-capacity flux leaves residual constraint. In ordinary low-flux settings, $|\boldsymbol{J}| \ll J_{\text{crit}}$ and the integral vanishes, so $M_s \approx 1$ and the vacuum appears memoryless. In extreme flux settings, a nonzero $M_s - 1$ would mean that the medium has not fully reset. This is a physical hypothesis, not a result already established by the present calculation.

This distinction matters for quantum decoherence. Within standard quantum mechanics, decoherence is described by entanglement with environmental degrees of freedom. CDFT can only add value if the proposed memory variable predicts a measurable residual phase, relaxation, or hysteresis effect beyond the standard description. Paper XII states this as an experimental target rather than as an assumed fact.

This is also the intended meaning of any observer-effect language in this release. An observer is not a disembodied human mind changing the vacuum by attention. Operationally, an observation is a physical coupling between the system and an apparatus, environment, detector, or record. If such coupling changes the state, the mechanism must be specified as an interaction channel. CDFT therefore treats "observation" as a boundary-and-measurement problem, not as a consciousness claim.

7.4 Operational dictionary relative to standard quantum language

The source notes for this release also proposed a comparison between the transport-medium vocabulary and standard quantum language. The professional way to use that comparison is not as a replacement of established theory, but as an operational dictionary: it states what CDFT would have to mean in measurable terms if the adaptive-vacuum interpretation is physical.

Feature	Standard reading	CDFT reading	Status
Vacuum	Background state with quantum fields	Effective transport medium with finite capacity C and response variables S, M_s	Hypothesis
Fine-structure constant	Empirical QED coupling	Equilibrium aspect ratio $\chi = R/a$ of the regulator vortex	Calibrated consistency check
Wave–particle duality	Complementary measurement descriptions	Smooth flux transport below threshold, vortex regulation near $\Psi_s \approx 1$	Interpretation
Decoherence	Environmental entanglement and phase suppression	Possible residual vacuum-memory lag in addition to standard decoherence	Test target
Planck constant	Fundamental quantum of action	Candidate product of a critical flux threshold and vortex period, $h = \Phi_c \tau_v$	Open derivation
Charge and mass	Empirical quantum numbers and inertial scale	Candidate boundary discontinuity and inertial resistance of a constrained vortex state	Open mapping
Gravity bridge	Spacetime curvature / Newtonian attraction	Possible secondary pressure-gradient limit of constrained transport	Speculative bridge

This table is deliberately conservative. It does not claim that CDFT has replaced QED or general relativity. It identifies the places where the transport-medium model must either produce new predictive value or fail.

7.5 Core-profile robustness program

The twelve-decimal recovery is an internal check of the stated equilibrium equation. It is not yet a proof that the same fixed point survives every possible vortex-core model. A professional robustness program must vary the internal vorticity profile and ask whether the same aspect ratio is recovered without retuning the transport stiffness:

Core model	Why it matters	Required test
Rankine / solid core	Baseline Lamb-style compact vortex model	Verify that the existing $\beta = 7/4$ calculation is stable under resolution and derivative checks.
Gaussian / diffuse core	Mimics smeared or soft-boundary flux density	Recompute β and the energy minimum with a width convention fixed before comparison.
Hollow or excluded core	Tests the extreme capacity-limited boundary case	Determine whether the back-pressure term changes functional form.
Finite-thickness vortex rings	Tests departure from the thin-ring approximation	Compare against finite-core ring families before claiming topological invariance.

Until those tests are implemented in the public supplementary code, the correct claim is conditional: the present model has a stable calibrated fixed point at $\chi = 1/\alpha$, and core-profile invariance is an open robustness target.

The current supplementary script now writes an explicit guardrail table and a matching figure under `outputs/paper_I/`. Those outputs do not prove invariance. They show how alternative core assumptions move the equilibrium when the Rankine calibration is held fixed, which is the correct starting point for the robustness program.

7.6 Connection to the lepton masses

This paper has focused entirely on the electron's aspect ratio. But the same vortex ring, when excited into three rotational modes related by 120° symmetry, produces a *trefoil* topology with $\mathbb{Z}_3$ symmetry. We show in Paper II that these three modes automatically satisfy the Koide mass formula:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3},$$

which holds experimentally to six significant figures. The electron, muon, and tau are not three different particles in CDFT. They are three orientations of the same vortex structure.

8 Necessary Conditions for the Model to Hold

The alpha recovery in this paper is conditional. The following conditions must hold for the CDFT interpretation to be more than a calibrated numerical coincidence.

1. **Topological resilience.** The aspect ratio $\chi = R/a$ must be a stable eigenmode, or fixed point, of the adaptive operating ratio Ψ_s. In practical terms, small perturbations of the ring geometry must relax back toward χ^* rather than drift to a continuum of nearby values. If perturbations select many unrelated aspect ratios with equal stability, the fine-structure interpretation fails.

2. **Flux–memory coupling.** The responsiveness S and memory M_s must be coupled so that the equilibrium vacuum satisfies $SM_s \to 1$ in the low-flux limit while still allowing a finite, measurable residual after overload. The simplest admissible behavior is an inverse correlation: if M_s rises after a high-flux event, the effective response must relax so that the attractor is restored instead of producing unbounded amplification.

3. **Non-radiative equilibrium.** The geometry $\chi^* \approx 137.035999177$ must represent a minimum of internal dissipation for the regulator vortex. In the language of this model, the equilibrium is a superfluid-like or zero-friction transport state; outside the minimum, excess curvature or excess ring size should introduce loss, radiation, or instability. This is a condition to be tested, not a claim already established by the twelve-decimal calculation.

4. **Independent stiffness closure.** The stiffness κ must be derived from the vacuum-medium properties without feeding in α, e, or $\hbar$ as hidden calibration targets. Until that is done, the Paper I result remains a high-precision consistency closure rather than a parameter-free derivation.

5. **External discriminant.** At least one measurable prediction must survive comparison with established physics. Candidate tests include vacuum hysteresis, extreme-field spectral residuals, decoherence-memory residuals, rotating-source pressure-gradient residuals, and transport-threshold tests in collective matter.

These conditions are intentionally strict. They are the difference between a suggestive geometric reconstruction of α and a physical theory that can be rejected by experiment.

9 Open Problems

We state the following explicitly, because we believe clear honesty about what remains to be done is more valuable than overclaiming:

1. **Derive κ from first principles (Paper III).** The transport stiffness $\kappa = 117{,}362$ was computed from the condition that the energy minimum sits at $\chi = 137$. It has not yet been derived independently from the vacuum medium properties ρ_0 and c. When that derivation is complete, the theory will predict α with no experimental constants as inputs.

2. **Derive $\hbar$ from circulation quantisation (Paper III).** We used $\Gamma = \hbar/m$ to derive R. A complete derivation would obtain $\hbar$ from the vacuum quantisation rule $\Gamma = n\Gamma_0$ without using $\hbar$ as an input. A concrete target suggested by the source notes is

$$h = \Phi_c \, \tau_v, \tag{22}$$

where Φ_c is a critical vacuum-flux threshold and τ_v is the characteristic period of the vortex attractor. At present this is a target identity for Paper III, not a completed derivation.

3. **Derive e from transport flux quantisation (Paper III).** We used $e^2/a = mc^2$ to derive a. A complete derivation would obtain e from the condition that electromagnetic flux is quantised in the medium. In the same language, charge would be the measurable discontinuity induced at the adaptive flux boundary rather than an unexplained label attached to the vortex.

4. **Test core-profile robustness without retuning κ.** The current public calculation uses the stated Lamb-style core model. A stronger derivation must repeat the stability calculation for hard, diffuse, hollow, and finite-thickness profiles with conventions fixed in advance.

5. **Measure or bound the vacuum-memory relaxation time τ_M.** The memory kernel in Eq. (21) is a hypothesis. It becomes physics only if an experiment or a deeper theory can identify, bound, or rule out the relaxation time and coupling strength.

6. **Find an external discriminant.** The recovery of $1/\alpha$ is a postdiction of an already known constant unless the framework also predicts a measurable effect not already accounted for by QED, general relativity, or standard condensed-matter and plasma models.

When the first-principles derivations and robustness tests are complete, $\alpha = 1/137.036$ would follow from the properties of the vacuum medium alone. Until then, the experimental value of α remains the calibration target used to test the internal consistency of the model.

10 Conclusion

We have presented two independent lines of evidence that the fine-structure constant α is the inverse aspect ratio of an equilibrium vortex ring in a constrained transport vacuum.

The first line: the vortex stability equation $dE_{\mathrm{norm}}/d\chi = 0$ has a valid, stable minimum at $\chi = 1/\alpha = 137.036$, recovered numerically to twelve decimal places. The equilibrium is physically natural — the two competing energy terms (circulation energy favouring large rings, transport back-pressure favouring small rings) balance with a ratio of 1.19, requiring no fine-tuning.

The second line: the vortex ring radius $R = \hbar/(mc)$, derived from quantum circulation quantisation, and the core radius $a = e^2/(mc^2)$, derived from electromagnetic self-energy, come from completely separate physics. Their ratio $R/a = \hbar c/e^2 = 1/\alpha$ agrees with the measured value to twelve decimal places.

Two different equations. Two different sectors of physics. One answer: 137.

The fine-structure constant is a shape.

Acknowledgments

The author used AI-assisted tools for formatting checks and code-verification support. The scientific claims, interpretations, and final manuscript decisions are the author's responsibility.

The author is indebted to the foundational work of Lord Kelvin [7] (vortex atom hypothesis), H. Lamb [8] (vortex ring energy), P.G. Saffman [9] (vortex dynamics), H. von Helmholtz [6] (vortex theorems), and P.A.M. Dirac [10] (vacuum as physical medium) — whose ideas this paper attempts to build upon.

A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in this Part I release archive. The broader public CDFD Runtime implements the same state grammar for reusable domain simulation, but the numerical values in this paper are reproduced from the paper-local supplementary script.

References

[1] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. *The Feynman Lectures on Physics*, volume 1. Addison-Wesley, 1964. Vol. 1, Ch. 4: "It has been a mystery ever since it was discovered more than fifty years ago, and all good theoretical physicists put this number up on their wall and worry about it.".

[2] Julian Schwinger. On quantum-electrodynamics and the magnetic moment of the electron. *Physical Review*, 73:416–417, 1948. First QED radiative correction $a_e = \alpha/(2\pi)$; establishes α as the perturbative expansion parameter of quantum electrodynamics.

[3] Arthur S. Eddington. On the value of the fine-structure constant. *Proceedings of the Royal Society of London A*, 122:358–369, 1929. Predicted $1/\alpha = 136$ from quantum combinatorics; incorrect value, no dynamical derivation.

[4] A. Wyler. L'espace symetrique du groupe des equations de Maxwell. *C. R. Acad. Sci. Paris, Serie A-B*, 269:A743–A745, 1969. Group-theory derivation giving $\alpha \approx 1/137.03$ — widely regarded as numerology; no physical mechanism proposed.

[5] Christian Beck and Michael C. Mackey. Electromagnetic dark energy. *International Journal of Modern Physics D*, 17(1):71–80, 2008. Stochastic electrodynamics approach to α; value not derived from first principles.

[6] Hermann von Helmholtz. Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen. *Journal für die reine und angewandte Mathematik*, 55:25–55, 1858. Founding paper of vortex dynamics; proves conservation of vortex tubes in inviscid flow.

[7] William Thomson. On vortex atoms. *Proceedings of the Royal Society of Edinburgh*, 6:94–105, 1869. Read April 25, 1867.

[8] Horace Lamb. *Hydrodynamics*. Cambridge University Press, 6th edition, 1932. Vortex ring energy formula, § 163; solid-rotation core gives $\beta = 1.75$.

[9] Philip G. Saffman. *Vortex Dynamics*. Cambridge University Press, 1992. Chapter 11: vortex ring self-induced velocity and energy.

[10] Paul A. M. Dirac. Is there an Aether? *Nature*, 168:906–907, 1951. Dirac argues the vacuum must be a physical medium with definite velocity at each point; reopens the aether question within quantum field theory.

[11] Maurizio Consoli and Lino Pappalardo. Emergent gravity and ether-drift experiments. General Relativity and Gravitation 42, 2585–2602, 2010. Vacuum condensate treated as a physical medium; light propagation in condensate background.

[12] John David Jackson. *Classical Electrodynamics*. Wiley, 3rd edition, 1999. Section 1.4: derivation of the classical electron radius $r_e = e^2/(m_e c^2) = 2.818 \times 10^{-15}$ m from electromagnetic self-energy condition.

[13] Peter J. Mohr, David B. Newell, Barry N. Taylor, and Eite Tiesinga. CODATA recommended values of the fundamental physical constants: 2022. *Reviews of Modern Physics*, 97:025002, 2025. $\alpha^{-1} = 137.035\,999\,177(21)$; NIST 2022 CODATA adjustment.